A GROUPOID AUTOMORPHISM ACTS TRIVIALLY ON THE CONORMAL SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; the author notes 'we will not need to know more about Y', keeping the chase self-contained. The author's label spelling 'idenity' is preserved verbatim.

Allowed standard prerequisites (editorial).

- Conormal sheaf and its functoriality. More on Morphisms of Spaces, lemmas conormal-functorial and conormal-functorial-more (cross-chapter, named). Fixes the canonical maps on C_{U0/U} whose compatibility with compositions is used.
- First-order neighbourhood data. equation-pull and the local structure section of this chapter (same chapter, structural anchors). Supplies the diagrams and the maps s, t, c whose conormal behaviour is compared.

Source reference. Original equation 03C7: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 303–303. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04CP: focused-seeds/geometry/S9-smg-idenity-on-conormal/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04G2: focused-seeds/geometry/S9-smg-idenity-on-conormal/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06BH: focused-seeds/geometry/S9-smg-idenity-on-conormal/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In the situation discussed in this section, let $\delta \in \Gamma_0$ and $f = t \circ \delta : U \to U$. If s, t are flat, then the canonical map $\mathcal{C}_{U_0/U} \to \mathcal{C}_{U_0/U}$ induced by f (More on Morphisms of Spaces, Lemma 04CP) is the identity map.*

Date: stacks-project:spaces-more-groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. To see this we extend the bottom of the diagram (03C7) as follows

$$\begin{array}{ccccc} Y & \longrightarrow & R \times_{s,U,t} R & \xrightarrow[\mathrm{pr}_0]{c} & R \xrightarrow{t} U \\ \downarrow & & \downarrow{\scriptstyle \mathrm{pr}_1} & & \downarrow{\scriptstyle s} \\ U & \xrightarrow[\delta]{} & R & \xrightarrow[t]{s} & U \end{array}$$

where the left square is cartesian and this is our definition of Y; we will not need to know more about Y. There is a similar diagram with similar properties obtained by base change to U_0 everywhere. We are trying to show that $\mathrm{id}_U = s \circ \delta$ and $f = t \circ \delta$ induce the same maps on conormal sheaves. Since s is flat and surjective, it suffices to prove the same thing for the two compositions $a, b : Y \to R$ along the top row. Observe that $a_0 = b_0$ and that one of a and b is an isomorphism as we know that $s \circ \delta$ is an isomorphism. Therefore the two morphisms $a, b : Y \to R$ are morphisms between algebraic spaces flat over U (via the morphism $t : R \to U$ and the morphism $t \circ a = t \circ b : Y \to U$). This implies what we want. Namely, by the compatibility with compositions in More on Morphisms of Spaces, Lemma 04G2 we conclude that both maps $a_0^*\mathcal{C}_{R_0/R} \to \mathcal{C}_{Y_0/Y}$ fit into a commutative diagram

$$\begin{array}{ccc} a_0^*\mathcal{C}_{R_0/R} & \longrightarrow & \mathcal{C}_{Y_0/Y} \\ \uparrow & & \uparrow \\ a_0^*t_0^*\mathcal{C}_{U_0/U} & = & (t_0 \circ a_0)^*\mathcal{C}_{U_0/U} \end{array}$$

whose vertical arrows are isomorphisms by More on Morphisms of Spaces, Lemma 06BH. Thus the lemma holds. □